


46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 08-02-26
Time: 3 Hrs Max. Marks: 180

## 08-02-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE ADV_RTA-1_SYLLABUS

**MATHEMATICS:** **PRESENT WEEK (100%): Functions, Limits, ITF, Continuity & Differentiability, Differentiation, AOD**

**PHYSICS:** **PRESENT WEEK (100%): Atoms, X-rays, Photoelectric effect, de-Broglie, Nuclei, KTG, Thermometry & Calorimetry, First law of thermodynamics**

**CHEMISTRY:** **PRESENT WEEK (100%): Solid State, Electrochemsitry, Alkyl Halides & Aryl Halides, Group-15**

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**TIME:3hrs** **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

## MATHEMATICS

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 1 – 4) | Questions with **Single Correct** Options | +3 | -1 | 4 | 12 |
| Sec – II (Q.N : 5 – 7) | Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0) | +4 | -2 | 3 | 12 |
| Sec – III (Q.N : 8 – 13) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +4 | 0 | 6 | 24 |
| Sec – IV (Q.N : 14 – 16) | Questions with **MATRIX MATCH** | +4 | -1 | 3 | 12 |
| **Total** | | | | **16** | **60** |

## PHYSICS

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 17 – 20) | Questions with **Single Correct** Options | +3 | -1 | 4 | 12 |
| Sec – II (Q.N : 21 – 23) | Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0) | +4 | -2 | 3 | 12 |
| Sec – III (Q.N : 24 – 29) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +4 | 0 | 6 | 24 |
| Sec – IV (Q.N : 30 – 32) | Questions with **MATRIX MATCH** | +4 | -1 | 3 | 12 |
| **Total** | | | | **16** | **60** |

## CHEMISTRY

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 33 – 36) | Questions with **Single Correct** Options | +3 | -1 | 4 | 12 |
| Sec – II (Q.N : 37 – 39) | Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0) | +4 | -2 | 3 | 12 |
| Sec – III (Q.N : 40 – 45) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +4 | 0 | 6 | 24 |
| Sec – IV (Q.N : 46 –48) | Questions with **MATRIX MATCH** | +4 | -1 | 3 | 12 |
| **Total** | | | | **16** | **60** |

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**MATHEMATICS** **MAX.MARKS: 60**

## SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks :** +3 If only the correct option is chosen.
**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)
**Negative Marks:** -1 In all other cases.

1. The number of points where $|xf(x)|+||x-2|-1|$ is non-differentiable in $x\in(0,3\pi)$,

where $f(x)=\prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^k}\right)}{3}\right)$ is _____

A) 4 B) 5 C) 7 D) 6

2. $y=\frac{\left(e^{\sin x}-1\right)^4+\left(\cos x-1\right)^3 e^{\sin x}+\left(e^x-e^{-x}\right)^3+x^3+2x^2+4}{e^x}$ then the value of $y^{11}(0)$ is

A) -4 B) 4 C) -8 D) 8

3. Let $\alpha_1,\alpha_2,\alpha_3,....\alpha_n$ be the distinct real values of $x$ satisfying the equation

$$\sin^{-1}\cos\left(\frac{2x^2+10|x|+4}{x^2+5|x|+3}\right)=\cot\left(\cot^{-1}\left(\frac{2-18|x|}{9|x|}\right)\right)+\frac{\pi}{2}$$

The absolute value of $\sum\alpha_1\alpha_2$

A) 4 B) 10 C) 8 D) 0

4. Let $I_1=\frac{\pi^2}{16}+\sqrt{2}$, $I_2=\left(\cot^{-1}\left(\frac{1}{\pi}\right)\right)^2+\frac{2\pi}{\sqrt{\pi^2+1}}$, $I_3=\left(\cot^{-1}\pi\right)^2+\frac{2}{\sqrt{\pi^2+1}}$, then which of the following is TRUE?

A) $I_1<I_2<I_3$ B) $I_1>I_2>I_3$ C) $I_2>I_3>I_1$ D) $I_2>I_1>I_3$



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## SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

**Full Marks :** +4 If only (all) the correct option(s) is (are) chosen.
**Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen.
**Partial Marks:** +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
**Partial Marks :** +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks:** -2 In all other cases.

5. Let $f_1(x)=\begin{cases}1 & , x\in(-\infty,0)\\ |x-1| & , x\in[0,3]\\ 2 & , x\in(3,8)\end{cases}$ and $f_2(x)=f_1(1-x)$, $f_3(x)=f_2(|x|)$, $f_4(x)=f_3(2x)+5$, then which of the following is/are incorrect?

A) $f_3(x)$ and $f_4(x)$ are even functions

B) $f_3(x)$ is many-one and onto function (Co-Domain of $f_3(x)$ is R)

C) Range of $[f_1(x)]\neq$ Range of $[f_2(x)]$ where [.] = greatest integer function

D) $f_1(x)+f_2(x)+f_3(x)+f_4(x)$ is an odd function

6. If $f:R\to[a,6]$ defined by $f(x)=\dfrac{x^2-2x+d}{x^2+3x+d}$ is an onto function, then

A) $a$ is equal to $\dfrac{2}{7}$ B) $d$ is equal to 4

C) $a$ is equal to 1 D) $d-7a$ is equal to $-3$

7. Let $p,q,r$ and $s$ are in A.P. and $f(x)=\begin{vmatrix}p+\sin x & q+\sin x & p-r+\sin x\\ q+\sin x & r+\sin x & -1+\sin x\\ r+\sin x & s+\sin x & s-q+\sin x\end{vmatrix}$ such that $\int_0^2 f(x)\,dx=-4$. If $d_1$ and $d_2$ be the possible values of the common difference of AP and $g(x)$ be the quadratic equation with $d_1$ and $d_2$ as the roots, then

A) The range of $g(x)$ is $[0,\infty)$

B) The number of points of non differentiability of the function $|g(x)|$ is 2

C) The number of points of non differentiability of the function $|g(|x|)|$ is 2

D) The number of solutions of $|g(|x|)|=|x|$ is 4



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## SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

8. If $f(x)=\sin^{-1}(x+2019)-\cos^{-1}(x+2021)$ and $g(x)=\sin^{-1}(x+2017)+\cos^{-1}(x+2019)$ and $x_1, x_2$ lies in domain of $f(x)$ & $g(x)$ respectively, then

$$\left|\lim_{x\to x_1+x_2}\frac{\sin(x-2x_1-2)-\sin(2x-3x_2+2022)}{\tan(x-x_1-x_2)}\right|$$ is equal to _____

9. If $2|x^2-4|=x^2-4x+a$ has exactly three distinct solutions then sum of all possible values of $3a$ is .....

10. Let the line $y=mx$ and $\log(x^2+y^2)=2\cot\sqrt{\frac{\pi^2}{16}-\alpha^2}\tan^{-1}\left(\frac{y}{x}\right)$ intersect each other at an acute angle $\theta$. If maximum value of $\theta$ is equal to $\frac{b\pi}{4}$, then $b$ is equal to

11. Let $f:R\to R$ be a function such that $f(x^5+x-2)=x^3-1\forall x\in R$ and $g$ be the inverse function of $f$. If $l=\frac{d}{dx}\left((x^3+1).g\left(f^3\left(g\left(f\left(g(x)\right)\right)\right)\right)\right|_{x=7^{\frac{1}{3}}}$ then value of $l\times 7^{-\frac{2}{3}}$ _____

12. Let $f(x)=\lim_{n\to\infty}\frac{x^{2n-1}+ax^3+bx^2}{x^{2n}+1}$ is continuous for all $x\in R$. If points $A(-a,3)$ and $B((b+1),-1)$ are points of local maximum and minimum of a cubic polynomial $y=g(x)$, then the value of $g(2)$ is:

13. Let $f$ be a differentiable function satisfying the functional rule $f(xy)=f(x)+f(y)+\frac{x+y-1}{xy}\forall x,y>0$ and $f'(1)=2$. The number of roots of the equation $f(x)=0$ is


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## SECTION-IV (Maximum Marks : 12)

This section contains **THREE (03)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 ONLYif the option corresponding to the correct combination is chosen;
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered);
**Negative Marks:** – 1 in all other cases.

14. Match the following

| | Column – I | | Column – II |
|---|---|---|---|
| I) | The number of solution(s) of the equation $\sin^{-1}\left(\frac{1}{1+x^2}\right)=\operatorname{sgn}\left(x^2+1\right)$, is/are | P) | 4 |
| II) | If the range of the function $f(x)=\cos^{-1}\left(\frac{x^2}{1+x^2}\right)$ is $(a,b]$, then $[a+b]$ is equal to (where [.] denotes greatest integer function) | Q) | 1 |
| III) | If $\alpha$ and $\beta$ are the roots of the $2x^2-3x-2=0$ then $\frac{12}{17}\left(\tan^2\left(\sin^{-1}\frac{\alpha}{\sqrt{1+a^2}}\right)+\tan^2\left(\cos^{-1}\frac{1}{\sqrt{1+\beta^2}}\right)\right)$ is | R) | 2 |
| IV) | If $f(x)=\sin\left(\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)+\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)$. Then the value of $\sum_{x=1}^{4} f\left(\frac{3x}{16}\right)$ is equal to | S) | 3 |
| | | T) | 0 |

A) $I\to P, II\to R, III\to S, IV\to R$ B) $I\to R, II\to Q, III\to S, IV\to P$

C) $I\to T, II\to P, III\to R, IV\to P$ D) $I\to Q, II\to P, III\to Q, IV\to R$

15. Match the following **List – I** with **List – II**

| | Column – I | | Column – II |
|---|---|---|---|
| P) | A continuous and differentiable function $f(x)$ satisfies functional equation $f(x+y)=\frac{f(x)f(y)}{f(x)+f(y)}$; $f(1)=16$, then $f(4)$ equals | 1) | 1 |
| Q) | Let $f:R\to R$ be a differentiable function satisfying $f\left(\frac{x+y}{5}\right)=\frac{6+f(x)+f(y)}{5}\forall\ x,y\in R$ and $f'(2)=2$, then area bounded by region $\lvert y\rvert=h(x)$, $x=2$ and $x=-2$ | 2) | 4 |



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| | | | |
|---|---|---|---|
| | where $h(x)=\left\|f(\|x\|)-4\right\|$ is | | |
| R) | Let $f$ be a differentiable function satisfying the relation $f(x+2y)=f(x)+f(2y)+e^{x+2y}(x+2y)-xe^{x}-2ye^{2y}+4xy$ $\forall x,y\in R$ and $f'(0)=1$, then the value of $\lim_{x\to 0}\frac{f(2x)-4x^2-2x}{x^2}$ is | 3) | 2 |
| S) | $f(x)=e^{x^3+x}$ and $g$ is inverse function of $f$. Then $\left\|g^{11}(1)\right\|=$ | 4) | 8 |
| | | 5) | 16 |

A) $P\to 5, Q\to 3, R\to 4, S\to 1$ B) $P\to 2, Q\to 4, R\to 2, S\to 1$

C) $P\to 4, Q\to 3, R\to 1, S\to 2$ D) $P\to 4, Q\to 3, R\to 2, S\to 1$

16. Match Column – I with Column – II

| **Column – I** | | **Column – II** | |
|---|---|---|---|
| I) | For $f(x)=\begin{cases}\frac{\|x-2\|}{x-2}[\log_e x], & x\in[1,3]-\{2\}\\ \{x^2\}, & x\in\left(3,\frac{7}{2}\right]\end{cases}$ number of points of discontinuity is (where [.] denotes GIF and {.} denotes fractional part function) | P) | 0 |
| II) | For $f(x)=\begin{cases}x^2 & if\ x\le 0\\ \sqrt{x} & if\ x>0\end{cases}$ number of points of inflection is | Q) | 5 |
| III) | Number of points where $f(x)=\max\left\{\left\|3-x^2\right\|,3-x^3\right\}$ is not differentiable | R) | 2 |
| IV) | The value of $2\lim_{x\to 0}x^2\left(1+2+3+....+\left[\frac{1}{\|x\|}\right]\right)$ is equal to Where [.] represents GIF | S) | 1 |
| | | T) | 3 |

A) $I\to Q, II\to S, III\to S, IV\to R$ B) $I\to T, II\to S, III\to P, IV\to S$

C) $I\to Q, II\to S, III\to R, IV\to S$ D) $I\to P, II\to Q, III\to R, IV\to S$



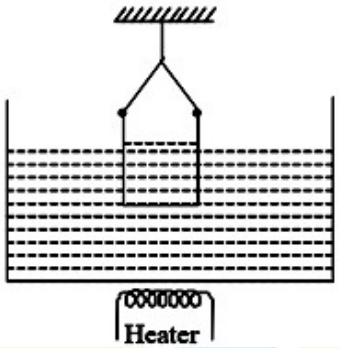
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**PHYSICS** **MAX.MARKS: 60**

## SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks :** +3 If only the correct option is chosen.
**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)
**Negative Marks:** -1 In all other cases.

17. Water is boiling in a large vessel as shown in the figure. Another open vessel containing water which is below $100^0 C$ is dipped in the bigger vessel by the help of string as shown in figure. Choose the correct option regarding the situation.



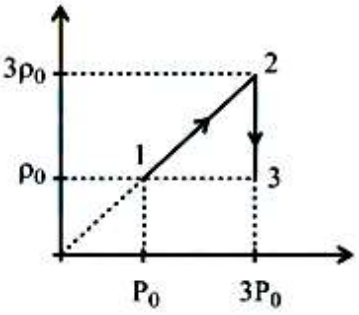

A) The temperature of water in the smaller vessel will gradually increase and it will start boiling below $100^0 C$

B) The temperature of water in the smaller vessel will gradually increase and it will start boiling above $100^0 C$

C) The temperature of water in the smaller vessel will gradually increase upto $100^0 C$ and it will not boil.

D) The temperature of water in the smaller vessel will gradually increase and it will start boiling at $100^0 C$

18. The density $\rho$ vs pressure P graph of an ideal monoatomic gas undergoes a process as shown in the figure. Find the total work done during the complete process. Molecular weight of gas is M and number of moles is 1

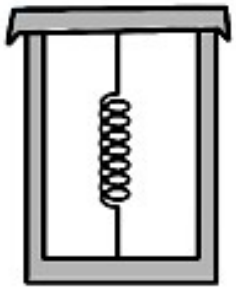



A) $\frac{P_0 M}{\rho_0}(1-\ln 3)$ B) $\frac{P_0 M}{\rho_0}(2-\ln 3)$ C) $\frac{P_0 M}{\rho_0}(3-\ln 3)$ D) None of these



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19. A container has a light lid held in place by a spring as shown in figure. Two moles of ideal mono atomic gas is kept inside the container having cross section area 100 $cm^2$ at 300 K at atmospheric pressure $10^5\,Pa$. The spring exerts a force of 1000 N when the lid is in place. The maximum amount of heat that can be given to the gas so that gas does not escape from the container. Assume heat capacity of the spring and container to be negligible. (R = universal gas constant in SI units).



A) 1200 R B) 900 R C) 1500 R D) 600 R

20. In a protonium atom, a proton and an antiproton orbit each other. What is the energy of a photon emitted in transition of a protonium atom from the first excited state to the ground state?

A) 5.1 eV B) 9.37 keV C) 12.5 keV D) 18.74 keV

## SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
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**Full Marks :** +4 If only (all) the correct option(s) is (are) chosen.
**Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen.
**Partial Marks:** +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
**Partial Marks :** +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks:** -2 In all other cases.

21. The pressure of an ideal gas is written by $P = K\frac{E}{V}$, where $E$ is some type of kinetic energy of random motion, $K$ is a constant and $V$ is the volume of gas. Consider the following statements and mark the correct option(s).

A) The gas is di-atomic, $K = \frac{2}{5}$ and $E =$ total kinetic energy

B) The gas is di-atomic, $K = 1$ and $E =$ rotational kinetic energy

C) The gas is monoatomic, $K = \frac{2}{3}$ and $E =$ total kinetic energy

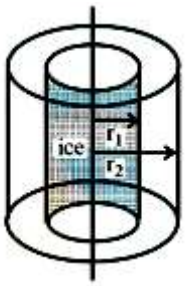
D) The gas is mono atomic or diatomic, $K = \frac{2}{3}$ and $E =$ translational kinetic energy.



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22. A 100 $cm$ long cylindrical flask with inner and outer diameter 2 $cm$ and 4 $cm$ respectively is completely filled with ice at $0^0C$ as shown in the figure. The constant temperature outside the flask is $40^0C$.

(Thermal conductivity of the flask is $0.693\,W/m^0C, L_{ice} = 80cal/gm$ & $\ell n2 = 0.693$, density of ice $= 900\,kg/m^3$).



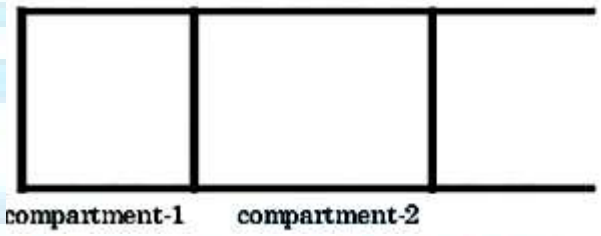

A) Rate of heat flow from outside to the flask is $80\pi\ J/s$.

B) The rate at which ice melts is $\frac{\pi}{4200} kg/s$

C) The rate at which ice melts is $100\pi\ kg/s$

D) Time taken for complete melting for ice is 378 sec.

23. Figure shows a long fixed container which has two freely movable (without friction) pistons. The container and pistons are made of a thermally conducting material that allows very slow transfer of heat. First compartment of container is filled with 2 moles of an ideal monoatomic gas at 200 K and the 2nd compartment is filled with 1 moles of ideal diatomic gas at 500 K. Initially pressure of gases in both the compartment is same and equal to atmospheric pressure. Temperature of atmosphere is 300 K. Finally, gases achieve equilibrium. 'R' is the gas constant




A) Heat transferred by gas in compartment – 2 to its surroundings is 700 R

B) Heat transferred by gas in compartment – 2 to the atmosphere is 500 R

C) Work done by gas in compartment – 2 on the gas in compartment – 1 is 200 R

D) Net heat transferred to gas in compartment – 1 is 500 R



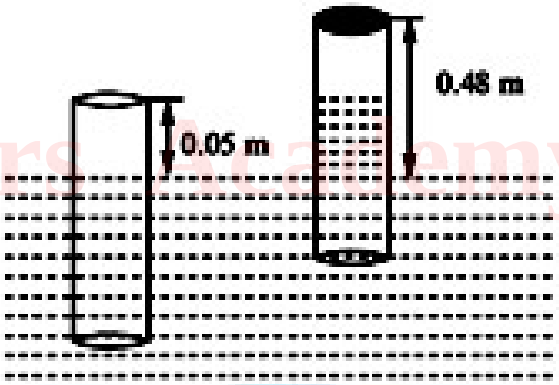
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## SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

24. A certain radio nuclide starts getting produced at a constant rate $10^4\,\text{sec}^{-1}$ by some mechanism at a certain moment $t=0$. At this moment there are already $10^5$ number of radio nuclides present in the sample. The mean life of this radio-nuclide is 20 seconds. At $t=40$ sec, the $\frac{dN}{dt}$ is $\frac{10^4}{K}$ (in $\text{sec}^{-1}$). Then find the value of $K$. (Take $e^2=7.4$)

25. An open end wide tube is immersed vertically in mercury in such a way that a length $0.05m$ extends above the mercury level. The open end of tube is then closed and the tube is raised slowly further by $0.43m$. The length of air column above the mercury level in the tube is _____ $cm$. $P_{atm}=(76cm)\rho_{Hg}\,g$. Ignore effect of surface tension.




26. A plane light wave of intensity $I=2kW/m^2$ falls on a plane mirror. The intensity of reflected light is 60% of intensity of the incident light. The angle of incidence is $30^0$. If the pressure exerted by the light on the mirror is $n\times10^{-6}\,N/m^2$, find the value of $n$

27. A container consists of a sample of hydrogen like ions of the same element & in the same state. In this state, the ionization energy of an ion in the above sample is 36% of a hydrogen atom. This sample is now allowed to reach its most stable configuration, thereby resulting in 10 distinct types of photons. The most energetic Paschen – line photons from this sample are made to incident on a metal. The stopping potential of the ejected photoelectrons is observed to be 4V. Find the work function (in eV) of the metal. (Take ionisation energy of a hydrogen atom $=13.6eV$ and truncate your final answer to two decimal places)


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28. Peak emission from a black body at a certain temperature occurs at a wavelength of $4000\overset{0}{A}$. On increasing its temperature, the total radiation emitted is increased to 256 times. These radiations are allowed to fall on metal surface. The photoelectrons emitted by peak emission wavelength at higher temperature can be brought to rest by applying a minimum potential which is equivalent to the excitation potential corresponding to the transition from level $n=2$ to level $n=4$ in the singly ionised Helium atom (obeys Bohr's Model). The work function of the metal (in eV) is

[Given E(in eV) = $\frac{12400}{\lambda(in\ \overset{0}{A})}$]

29. Consider the following nuclear fusion reaction

$${}_1^2H + {}_1^2H \rightarrow {}_1^3H + {}_1^1P$$

Assume binding energy per nucleon of deuterium and tritium are $1\ MeV$ and $2.80\ MeV$ respectively. Neglect the kinetic energy of the nuclei before the fusion. Find the kinetic energy (in MeV) of the tritium produced in the above nuclear reaction.

## SECTION-IV (Maximum Marks : 12)

This section contains **THREE (03)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 ONLYif the option corresponding to the correct combination is chosen;
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered);
**Negative Marks:** – 1 in all other cases.

30. The ground state energy of hydrogen atom is -13.6 eV. A sample of hydrogen atoms in its ground state is irradiated with photons of energy 10.2 eV. The radiations from the above sample is then incident on two sample containing excited $He^+$ ions and excited $Li^{2+}$ ions respectively. Both the samples absorb the radiations. (use $hc = 12400\ eV\ \overset{o}{A}$)



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| List – I | | List – II | |
|---|---|---|---|
| I) | No. of spectral lines obtained in the Spectra of $Li^{2+}$ | P) | 6 |
| II) | No. of spectral lines obtained in the Spectra of $He^{+}$ | Q) | 15 |
| III) | Smallest wavelength in spectra of $He^{+}$ is $(48.8)X\,\overset{o}{A}$ then X (Approx.) is | R) | 5 |
| IV) | Smallest wavelength in spectra of $Li^{2+}$ is $(10.4)Y\,\overset{o}{A}$ then Y (Approx.) is | S) | 10 |
| | | T) | 18 |
| | | | |

Codes:

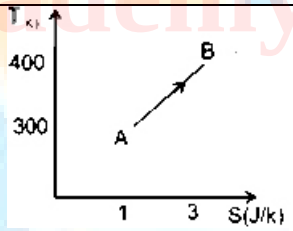
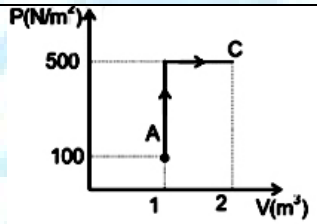
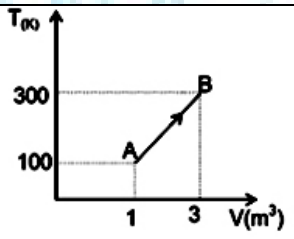
A) I – R; II – P; III – P; IV – T

B) I – Q; II – P; III – P; IV – Q

C) I – Q; II – P; III – R; IV – S

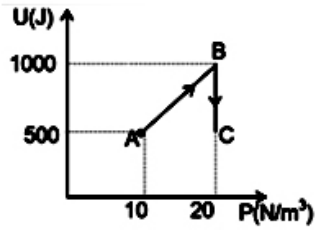
D) I – T; II – P; III – R; IV – Q

31. In **List – I**, difference thermodynamics process are given and in **List – II** magnitude of possible heat exchange between system and surrounding are given due the process.

| List – I | | | List – II | |
|---|---|---|---|---|
| I) | 2mole monoatomic ideal gas is taken through process $A \to B$ which is shown in temperature (T) and entropy(s) diagram |  | P) | 1850J |
| II) | 2 mole ideal monoatomic gas is taken through process $A \to B \to C$ as shown in P-V diagram |  | Q) | 8314J |
| III) | Two mole monoatomic ideal gas is taken through the process $A \to B$ as shown in the T-V diagram |  | R) | 333J |



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| | | | | |
|---|---|---|---|---|
| IV) | One mole monoatomic ideal gas is taken through the process ABC as shown in the figure, with the help of internal energy U and pressure P of the gas |  | S) | 400J |
| | | | T) | 700J |

Which one of the following option is correct?

A) $I \to T, II \to P, III \to R, IV \to Q$ B) $I \to T, II \to R, III \to P, IV \to Q$

C) $I \to T, II \to P, III \to Q, IV \to R$ D) $I \to Q, II \to P, III \to Q, IV \to S$

32. List – I shows various functional dependencies of energy (E) on the atomic number (Z). Energies associated with certain phenomena are given in List – II.

Choose the option that describes the correct match between the entries in List – I to those in List – II

| List – I | | List – II | |
|---|---|---|---|
| P) | $E \propto Z^2$ | 1) | Energy of characteristic x-rays |
| Q) | $E \propto (Z-1)^2$ | 2) | Electrostatic part of the nuclear binding energy for stable nuclei with mass numbers in the range 30 to 170 |
| R) | $E \propto Z(Z-1)$ | 3) | Energy of continuous x-rays |
| S) | E is practically independent of Z | 4) | Average nuclear binding energy per nucleon for stable nuclei with mass number in the range 30 to 170 |
| | | 5) | Energy of radiation due to electronic transitions from hydrogen-like atoms |

A) $P \to 4, Q \to 3, R-1, S \to 2$ B) $P \to 5, Q \to 2, R-1, S \to 4$

C) $P \to 5, Q \to 1, R-2, S \to 4$ D) $P \to 3, Q \to 2, R-1, S \to 5$



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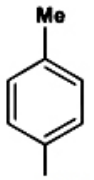
**CHEMISTRY** **MAX.MARKS: 60**

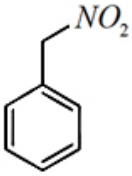
## SECTION–II (Maximum Marks: 12)

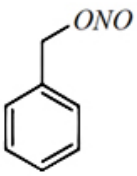
This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks :** +3 If only the correct option is chosen.
**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)
**Negative Marks:** -1 In all other cases.

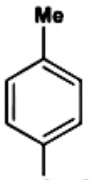
33. The major product formed in the following reaction is:

$$\text{n-heptane} \xrightarrow[773K\,10atm]{\substack{1.Cr_2O_3-Al_2O_3 \\ 2.NBS/CCl_4,hv \\ 3.AgNO_2}} [P]$$

A) 


B) 


C) 


D) 


34. $CaCO_3 \xrightarrow{\Delta} A + CO_2;$

$A + coke \xrightarrow{2773K,\,Electric\ furnace} B + CO;$

$B + N_2 \xrightarrow{1273K} D + \text{graphite}\ ;$

$D \xrightarrow{steam,\,473K} E + other\ \ product\ .$

**Information: 'E' contain 'N' atom**

Which of the following is **incorrect** about the compounds A,B,D,E ?

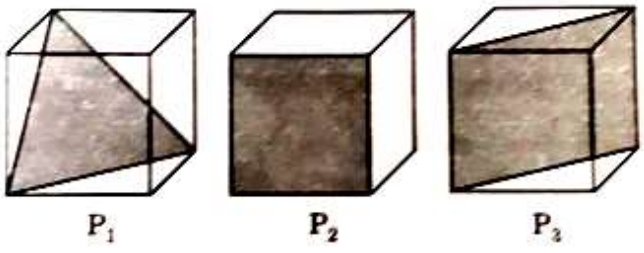
A) The anion of compound 'B' contain triple bond.

B) The compound 'D' can be used as fertilizer.

C) The compound 'E' can be dried with anhydrous $CaCl_2$.

D) The compound 'E' can also be obtained by the hydrolysis of $Mg_3N_2$.

35. Following three planes $(P_1, P_2, P_3)$ in an FCC unit cell are shown. Consider the following statements and choose the correct option that follow:






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**(P)** $P_1$ contains no voids three dimensions

(Q) $P_2$ contains only octahedral voids.

(R) $P_3$ contains both octahedral and tetrahedral voids.

A) All are true

B) Only $P$ and $Q$ are true

C) P and R are true

D) Only R is true

36. Let a fully charged lead-storage battery contains 2.0 L of $4M$ $H_2SO_4$. What will be the concentration of $H_2SO_4$ in the battery after 1.93 ampere current is drawn from the battery for 20 hours ? (Assume no change in the volume of acid in the battery)

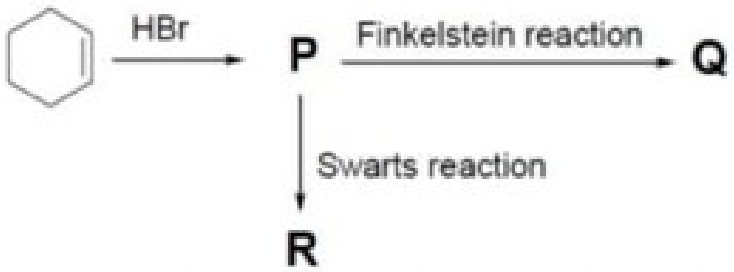
A) 9.67 M B) 1.64 M C) 6.56 M D) 3.28M

## SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

**Full Marks :** +4 If only (all) the correct option(s) is (are) chosen.
**Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen.
**Partial Marks:** +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
**Partial Marks :** +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks:** -2 In all other cases.

37. For the reaction sequence given below, the incorrect statement(s) is(are)




(In the options, X is any atom other than carbon and hydrogen, and it is different in P, Q and R)

A) C-X bond length in P, Q and R follows the order Q > R > P

B) C-X bond enthalpy in P, Q and R follows the order R > P > Q

C) Relative reactivity toward $S_N2$ reaction in P, Q and R follows the order P > R > Q

D) $pK_a$ value of the conjugate acids of the leaving groups in P, Q and R follows the order R > Q > P



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38. In which of the following reaction(s) $POCl_3$ is obtained as one of the product?

A) $PCl_5 + P_4O_{10} \xrightarrow{(6:1)}$

B) $PCl_5 + H_2O \xrightarrow{(1:1)}$

C) $PCl_5 + SO_2 \xrightarrow{(1:1)}$

D) $PCl_3 + C_2H_5OH \xrightarrow{(1:3)}$

39. The standard reduction potentials of some half-cell reactions are given below:

$PbO_2 + 4H^+ + 2e^- \rightleftharpoons Pb^{+2} + 2H_2O;\ E^0 = +1.455V$

$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{+2} + 4H_2O;\ E^0 = +1.51V$

$Ce^{+4} + e^- \rightleftharpoons Ce^{+3};\ E^0 = +1.61V$

$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O;\ E^0 = +1.71V$

Pick out the correct statement: ()

A) $PbO_2$ will oxidise $Mn^{+2}$ to $MnO_4^-$

B) $MnO_4^-$ will oxidise $Pb^{+2}$ to $PbO_2$

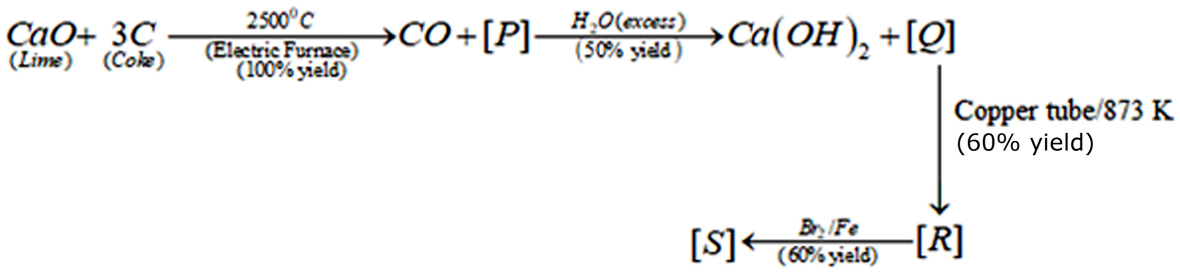
C) $H_2O_2$ will oxidise $Mn^{+2}$ to $MnO_4^-$

D) $Ce^{+4}$ will oxidise $Pb^{+2}$ to $PbO_2$

## SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

40. The weight of $[S]$ (in g) produced if 5.6 kg of lime is used in the following reaction sequence is _____ in gms.

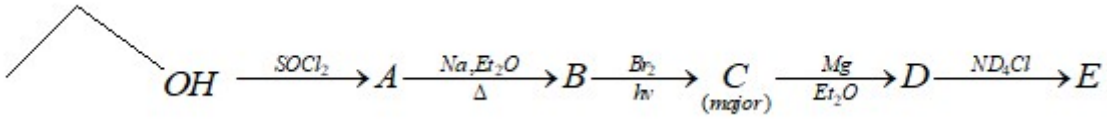



(Given: Atomic weight H = 1, C = 12, O = 16, Br = 80, Ca = 40)

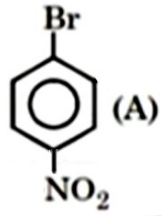


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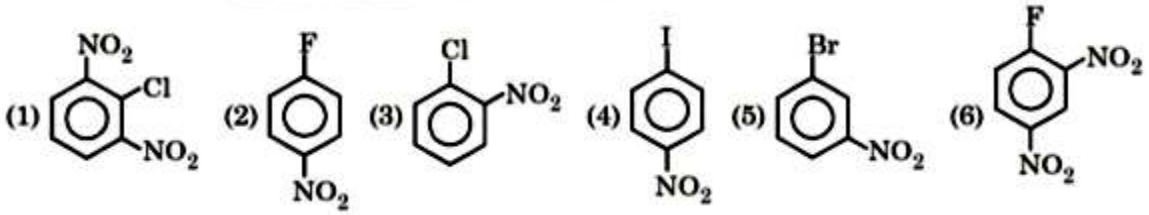
41.

a) 


X = Total no. of possible monochloro derivatives $(C_4H_8DCl)$ of E excluding stereo isomers. The value of X is _____

b) 


Number of compounds which can show faster rate of nucleophilic substitution of halogens than compound (A) in the given below compounds is Y




Then find the $X + Y =$ _____

42. Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance.

(i) Remove all the anions (X) except the central one

(ii) Replace all the face centered cations (M) by anions (X)

(iii) Remove all the corner cations (M)

(iv) Replace the central anion (X) with cation (M)

The value of $\left(\frac{number\ of\ anions}{number\ of\ cations}\right)$ in Z is _____

43. Ionic compound AB have $NaCl$ type structure and the density is $1.6\,gm/cm^3$ . 6 grams of this compound is added in 200 gm of water. Assume 100% ionization of AB in solution. The freezing point of the solution is 269.28K. If radius of anion is 170 pm, then calculate the nearest integer value of 'X',

Where $X = \left(\frac{radius\ of\ anion}{radius\ of\ cation}\right) \times 8$;

**[Given,** $K_f(H_2O) = 1.86\,kg\,mol^{-1}$, $N_A = 6 \times 10^{23}$

**Normal freezing point of water = 273K]**



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44. 0.1M aqueous solution of HCOOH shows a resistance of $50\Omega$ filled in a cell having separation between parallel electrodes 4 cm and cross section area of electrode $10cm^2$. The value of dissociation constant of the acid in scientific notation is $\left(x\times10^{-y}\right)$. The value of $(x+y)$ is **(nearest integer)**

Given $\Lambda_m^\infty[Ca(HCOO)_2]=230Scm^2mol^{-1}$

$\Lambda_m^\infty[CaCl_2]=280Scm^2mol^{-1}$; $\Lambda_m^\infty[HCl]=425Scm^2mol^{-1}$

45. If EMF of following cell:

$$Pt_{(s)}\left|\underset{1M}{Fe^{2+}_{(aq)}},\underset{1M}{Fe^{3+}_{(aq)}}\right\|\left|\underset{1M}{MnO^-_{4(aq)}},\underset{1M}{Mn^{2+}_{(aq)}},\underset{pH=5}{H^+_{(aq)}}\right|Pt_{(s)}$$ is $\left(y\times10^{-2}\right)$ volts, then the value of '$y$' is

(Given: $E^\circ_{MnO_4^-|Mn^{2+}}=+1.51V$; $E^\circ_{Fe^{3+}|Fe^{2+}}=+0.78V$, $\dfrac{2.303RT}{F}=0.06$)

## SECTION-IV (Maximum Marks : 12)

This section contains **THREE (03)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 ONLYif the option corresponding to the correct combination is chosen;
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered);
**Negative Marks:** – 1 in all other cases.

46. Match each reaction in List – I with the feature of its major product in List – II and choose the correct option.

| List – I | | List – II | |
|---|---|---|---|
| P) | 1-chloro-3-methylpentane $\xrightarrow[S_N2\text{ reaction}]{aq.NaSH}$ (single stereoisomer) | 1) | Inversion of configuration |
| Q) | 2-bromobutane $\xrightarrow[S_N2\text{ reaction}]{aq.NaSH}$ (single stereoisomer) | 2) | Retention of configuration |
| R) | 5-bromo-2,3-dimethylhexane $\xrightarrow[S_N1\text{ reaction}]{aq.NaSH}$ (single stereoisomer) | 3) | Mixture of enantiomers |
| S) | 3-chloro-2-phenylpentane $\xrightarrow[S_N1\text{ reaction}]{aq.NaSH}$ (single stereoisomer) | 4) | Mixture of structural isomers |
| | | 5) | Mixture of diastereomers |

A) $P\to1; Q\to1; R\to3; S\to3$ B) $P\to2; Q\to1; R\to5; S\to4$

C) $P\to4; Q\to2; R\to5; S\to3$ D) $P\to2; Q\to1; R\to5; S\to3$



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47. Match the items of Column – I & Column – II

| | Column – I | | Column – II |
|---|---|---|---|
| P) | Leclanche Cell | 1) | Cell Reaction: $2H_2 + O_2 \rightarrow 2H_2O$ |
| Q) | Ni – Cd Cell | 2) | Does not involve any ion in solution and is used in hearing aids |
| R) | Fuel Cell | 3) | Rechargeable |
| S) | Mercury Cell | 4) | Reaction at anode: $Zn \rightarrow Zn^{2+} + 2e$ |
| | | 5) | Converts energy of combustion into electrical energy |

A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 2$ B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1,5; S \rightarrow 2$

C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$ D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 3$

48. Match the following:

| Column – I | | Column – II | |
|---|---|---|---|
| P) | $P_2O_3 + 3H_2O \rightarrow$ | 1) | One of the product on hydrolysis give a tribasic acid. |
| Q) | $P_4 + 3NaOH + 3H_2O \rightarrow$ | 2) | One of the product is monobasic acid |
| R) | $PCl_5 + CH_3COOH \rightarrow$ | 3) | One of the product is a dibasic acid |
| S) | $H_3PO_2 + 2H_2O + 4AgNO_3 \rightarrow$ | 4) | One of the product can act as reducing agent. |
| | | 5) | In one of the product the central atom undergoes $sp^3$ hybridisation. |

A) $P \rightarrow 3,4,5; Q \rightarrow 4,5; R \rightarrow 1,2,4,5; S \rightarrow 2,4,5$

B) $P \rightarrow 3,4,5; Q \rightarrow 1,4,5; R \rightarrow 1,2,4,3; S \rightarrow 2,3,5$

C) $P \rightarrow 3,4,5; Q \rightarrow 4,5; R \rightarrow 1,3,4,5; S \rightarrow 2,3,4,5$

D) $P \rightarrow 1,2,4,5; Q \rightarrow 3,4,5; R \rightarrow 4,5; S \rightarrow 2,3,4,5$



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